HDD Market & Competitor Report

Demand drivers, market financials, competing technologies and a 2026–2031 value forecast
Report 1 of 3 · Prepared 28 October 2026

~$36B	~2.0 ZB	~58%	2028–2031
2026E HDD industry revenue (vs $23.3B in 2025)	2026E exabytes shipped (+25–30% YoY)	Seagate / WD gross margin Sep-2026 quarter (record)	Long-term supply agreements already signed out to

Bottom line

HDD is not a dying business — it is the bulk-capacity layer of the AI data center, and it is in the strongest period of its history. Demand measured in exabytes is contracted through 2028 and, for some customers, to 2030–31. Units are flat (~125M/yr) while capacity per drive and price per TB rise. The 5-year risk is **cyclical, not technological**: an AI-capex digestion phase and a NAND oversupply around 2028–29 could compress prices and margins, but no competing medium can absorb HDD's ~2 ZB/yr of shipments at comparable cost before 2031.

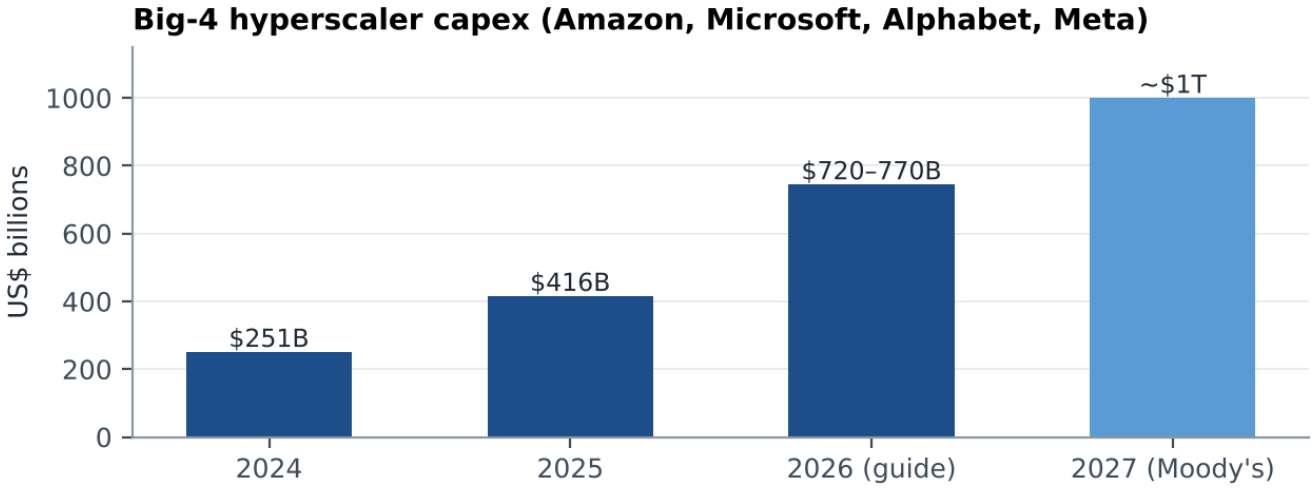
Key findings

- **Macro driver:** Big-4 hyperscaler capex went ~$251B (2024) → ~$416B (2025) → ~$720–770B guided for 2026; Moody's sees ~$1T in 2027. AI training data, checkpoints, inference outputs and agent traces must be retained, and ~80–90% of hyperscale data sits on HDD.
- **HDD market:** industry revenue $19.8B (2024) → $23.3B (2025) → ~$36B (2026E). Price per TB is *rising* (WD +28% YoY) for the first time in decades. Nearline is ~88–90% of exabytes and ~80–88% of revenue.
- **Main competitor — enterprise QLC SSD:** NAND is in an even bigger supercycle (Micron 87% gross margin, top-5 eSSD revenue $37.6B in 2Q26). But SSD $/TB rose ~6x in a year, so the SSD:HDD price gap widened from ~5–6x to ~16–23x. The shortage *helped* HDD.
- **Technology:** HAMR is in volume (Seagate 44TB since Mar-2026; WD 44TB HAMR in 1H27; Toshiba ~2028). Roadmaps point to 50–60TB in 2028 and ~100TB around 2029–31 — enough to grow exabytes ~20–25%/yr with flat unit output.
- **Forecast (my base case):** HDD revenue ~$54B by 2031 (~8% CAGR from 2026), exabytes ~5 ZB. Bear case ~$35B (flat), bull case ~$72B. Exabyte growth is near-certain; dollar growth after 2027 depends on pricing discipline.

Method: eight research agents gathered 2025–2026 data from SEC filings, earnings calls, TrendForce, Gartner, Coughlin Associates, Furthur Market Research, Blocks & Files and company roadmaps. Items marked **[EST]** are analyst estimates. Not investment advice.

1. The tech market and what it means for HDD

The single most important driver of HDD demand today is AI-related data-center construction. Most AI capex buys GPUs, networking, power and buildings — storage is a single-digit share — but every cluster creates data that has to be ingested and kept: training corpora, checkpoints, logs, evaluation sets, and (increasingly) inference outputs and agent traces retained for compliance and re-training.



Sources: company filings via Platformonomics; Moody's via DatacenterDynamics. Citi projects $5.35T cumulative 2027–2030.

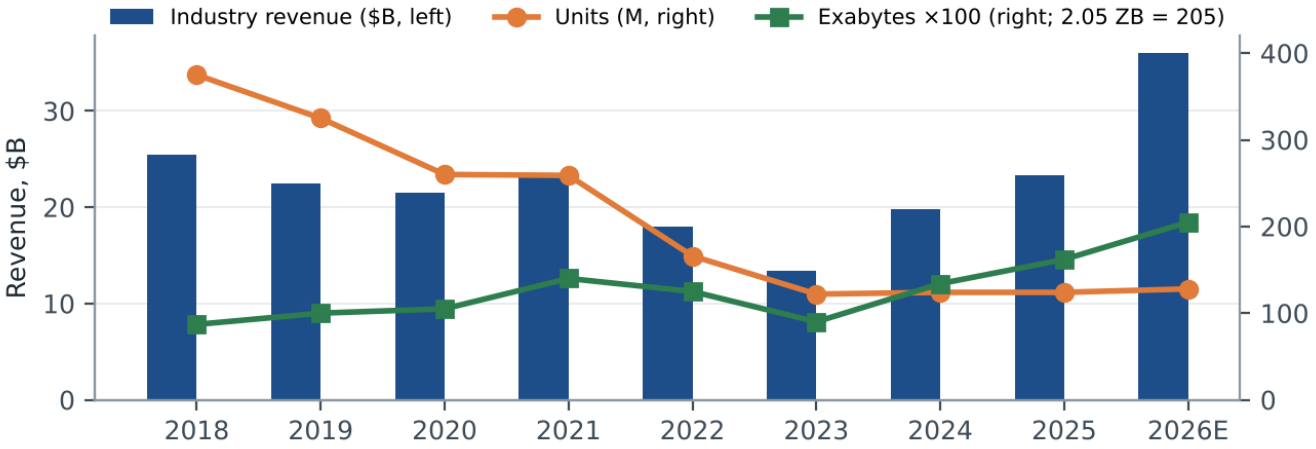
Driver	Latest data point	Effect on HDD
Hyperscaler capex	~$720–770B in 2026; ~$1T 2027 (Moody's)	Strong, 1–3 yr lag; drives nearline exabytes
Cloud backlogs	AWS $496B, Google Cloud $514B, Microsoft RPO $678B	Supports at least 2 more years of build-out
Gartner IT spend 2026	$6.37T (+14.2%); data-center systems $822B (+62.5%)	DC growing 4–5x faster than IT overall
Devices (PCs, phones)	+9.8% in 2026	Legacy HDD (PC/consumer) keeps shrinking
Data creation (IDC)	~20–25% CAGR [approx.]; only ~2% is stored	Retention policy, not creation, decides HDD demand
Edge / physical AI	Seagate Edge-IoT +52% YoY (Sep-26 qtr)	Small share of exabytes; NAS/surveillance support

How the demand flows

- **AI training** → data lakes in object storage (S3 / GCS / Azure Blob) → nearline HDD as the bulk tier.
- **Inference and agents** → prompts, outputs, traces and generated media → retained on HDD for audit and re-training.
- **Hot data** → QLC SSD (GPU feeding). HDD keeps the 70–80% of data that is not hot.
- **Customer base is widening** from the top hyperscalers to neoclouds, AI labs, sovereign clouds and enterprise.

2. HDD market financials

Units collapsed from the PC era (~375M in 2018, ~600M+ at the 2010 peak) to ~124M and have now flattened. Exabytes and revenue have decoupled from units: capacity per drive has gone from ~2.3TB to ~16TB, and in 2025–26 price per TB started rising.



2018–2022 values approximate; 2023 derived; 2024–2025 Coughlin Associates; 2026E derived from Seagate/WD reported and guided quarters plus a Toshiba estimate.

Market segments

Segment	Share of exabytes	Share of revenue	Trend to 2031
Nearline / mass capacity (cloud + enterprise, 3.5" 20–44TB)	~88–90%	~80–88%	Growing; the only engine
PC / client (2.5" and 3.5")	<3%	~5%	Structural decline (SSD boot drives)
Consumer external / NAS	~3–5%	~4–6%	Flat to down
Surveillance / video / CE	~4–6%	~4–6%	Flat; edge AI helps, China weak
Mission-critical 10K/15K	<1%	~1–2%	End of life (replaced by SSD)

Profitability and cost structure (Seagate and WD as proxy)

Metric	Seagate FY26	Seagate Sep-26 qtr	WD FY26	WD Sep-26 qtr	WD Dec-26 guide
Revenue	$12.20B	$4.23B (+61%)	$12.92B (+36%)	$4.38B (+55%)	$5.1B
Gross margin (non-GAAP)	45.6% (GAAP)	57.8%	49.1%	58.1%	60–61%
Operating margin (non-GAAP)	36.5%	50.7%	37.3%	49.1%	~52% [EST]
Opex per quarter	~$300M	~7% of revenue	—	$391M (~9%)	$400–410M
Capex	4.7% of revenue	$211M (5%)	4–6%	4–6%	4–6%

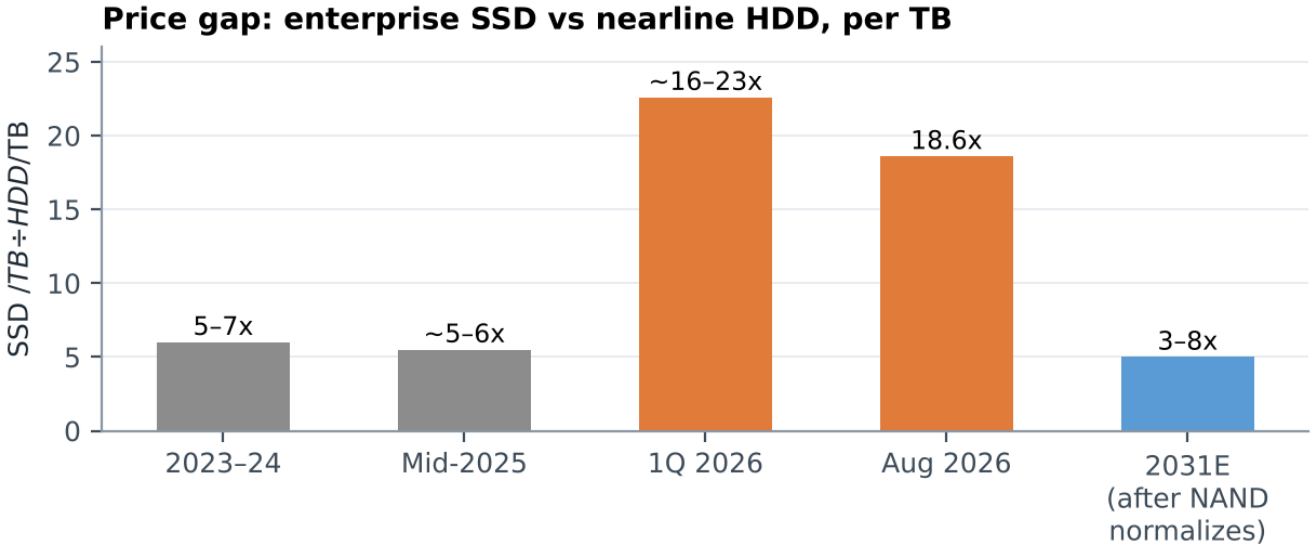
- **Margins are at all-time highs** — well above the 2018–19 peak (~30–33% gross margin) and the 2023 trough (~18–25%). Price per TB is up 20–30% while cost per TB falls ~5–8%; that spread explains the jump.

- **Capex is only 4–6% of revenue.** Exabyte growth comes from areal density, not new factories. Neither Seagate nor WD is adding unit capacity. Toshiba is the exception (~$380M to double Philippine output by FY2027).
- **Supply is contracted:** nearline lead times exceeded 52 weeks in 2025. Seagate has most supply under long-term agreements into CY2028 (some to 2030), with prices fixed ~12 months ahead; WD has visibility through 2030–31 with its top 7 customers.

3. Competitors: SSD/NAND, tape and new media

The real competitor is high-capacity enterprise QLC SSD (122TB shipping, 245–256TB arriving, 512TB planned for 2027). All-flash vendors — Everpure (formerly Pure Storage) claims no new HDDs will be sold after 2028, and VAST Data is valued at $30B — market directly against nearline HDD. Tape and new archive media only compete at the cold-archive edge.

Competitor	Scale (latest)	Margin / economics	Threat to nearline HDD by 2031
NAND / enterprise SSD (Samsung, SK hynix-Solidigm, Micron, Kioxia, Sandisk, YMTC)	Top-5 NAND revenue $68.9B in 2Q26 (+77% QoQ); eSSD $37.6B in 2Q26; 2026E NAND ~$300B+	Micron 87% GM; Sandisk 84.6% GM; NAND capex only $22.2B (+5%) in 2026	**Medium.** Takes hot/warm and power-limited tiers; may reach ~20–30% of DC capacity by 2030 [EST]
Tape (LTO: IBM, HPE, Quantum; media Fujifilm, Sony)	160 EB (compressed) in 2025, ~3% of HDD exabytes; LTO-10 40TB	Media <$1–5/TB; slow access	**Low.** Archive only
Glass / ceramic / DNA (Project Silica, Cerabyte, Atlas)	Pilots; Cerabyte 1PB rack 2026 → 100PB/rack 2030	Not yet commercial	**Very low** before 2030



Sources: Micron FMS 2025; VDURA Flash Volatility Index via Blocks & Files and Notebookcheck; 2031 range is an analyst estimate.

Did the NAND shortage help HDD? Yes — strongly.

- Mid-2025: AI demand outran HDD supply; lead times passed 52 weeks. Hyperscalers (reportedly Microsoft, Meta) shifted orders to QLC SSD.
- That extra demand landed on a NAND industry with almost no new capacity. A 30TB QLC SSD went from ~$2,450 to ~$15,100; HDD prices rose only ~35%.

- SSD lost its cost case for bulk storage; HDD makers sold out and signed multi-year contracts. Seagate gross margin went from 40% to 57.8% in a year.
- **Risk ahead:** new NAND fabs arrive late 2027–2028 and contract price floors expire. A NAND glut around 2028–29 could shrink the gap back to ~3–5x and restart the substitution push.

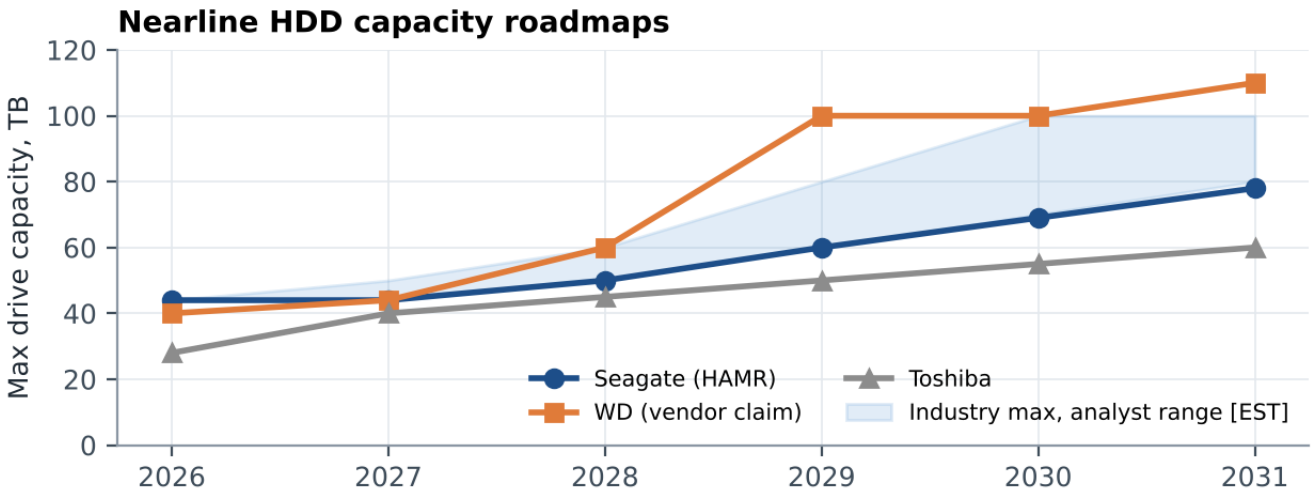
TCO argument, both sides

Flash camp says	HDD camp says
SSD is ~10x denser (TB per cm³); one 245TB module replaces 8–10 HDDs	Purchase price dominates TCO; QLC's power saving is worth <$1/TB over 5 years
~30% less power per TB stored (TrendForce); longer life, fewer failures	Replacing ~2 ZB/yr of HDD would need several hundred billion $ of NAND fabs; NAND capex is ~$22B/yr
Pure/Everpure: no new HDDs after 2028; real hyperscale win ('tens of EB' into 2028)	No independent analyst supports 2028; Monroe, Coughlin and Micron all see HDD exabytes growing to ~2030–31

4. Technology: HDD roadmap vs competitors

2026–2031 is the strongest HDD technology cadence since helium. HAMR provides a credible path from ~3TB to ~10TB per platter (~3.3x), platter count adds 10 → 14 (~1.4x), and multi-actuator / multi-head designs attack HDD's weakest metric, IOPS per TB.

Technology	What it does	Status Oct-2026	Impact 2026–2031
HAMR (heat-assisted)	Laser + near-field transducer heat FePt media during write	Seagate Mozaic 3+/4+ in volume (44TB); WD 1H27; Toshiba ~FY28	**Main capacity engine** to ~100TB
ePMR + UltraSMR (WD)	Bias-current-assisted write; shingled tracks +15–20%	WD 40TB shipping since mid-2026	Bridge to 60TB (2028) at flat power
MAMR / FC-MAMR (Toshiba)	Microwave/flux-assisted write	24TB CMR / 28TB SMR	Toshiba's bridge to 40TB (2027)
More platters	10 → 11 → 12 (glass) → 14	WD 11; Toshiba 12 verified; WD 14 planned	~1.4x capacity; harder mechanics
Multi-actuator / HBD	Two actuators (MACH.2, Dual Pivot) or several heads active at once	MACH.2 small volume; WD HBD sampling; Dual Pivot 2028	Restores IOPS/TB; defends vs QLC
OptiNAND, TDMR, triple-stage actuation	On-drive flash metadata; 2D readers; VCM+PZT+thermal	In production / on roadmap	Enable ~800–1,000 kTPI
HDMR / bit-patterned media	HAMR on discrete magnetic dots	Research	2030s, beyond 100TB



Vendor roadmaps (Seagate, WD Innovation Day Feb-2026, Toshiba slides via TechRadar). Expect 6–18 months of slip on the boldest points.

Competitor technology

- **NAND:** layer counts passing 300–400 (Samsung V10 400+, SK hynix 375, Kioxia BiCS10 332). QLC mainstream; PLC has no date. Cost per bit falls ~15–25%/yr historically, but each new layer generation adds etch, deposition and bonding cost.
- **Tape:** LTO-10 40TB native; roadmap was *cut* in Nov-2025 (LTO-14 now 365TB, from 576TB).
- **Glass / ceramic / DNA:** archive-only, write-once; commercial scale ~2030 at best.
- **Verdict [EST]:** no competitor closes the $/TB gap with nearline HDD by 2031. Expect ~5–8x at device level once NAND normalizes and ~3–5x on full TCO for read-heavy workloads. HDD keeps ~80–90% of hyperscale stored exabytes through 2031. The bigger threat is after 2031, and depends on whether HDD's parallelism architectures succeed.

Why this matters for a servo engineer: almost every remaining HDD scaling lever is a mechatronics problem — ~25nm track pitch (2.5–3nm 3σ position error), 12–14 thinner glass platters (flutter, lower arm modes), triple-stage actuation, dual-pivot cross-coupling, multi-head simultaneous writing, and HAMR laser/protrusion transients. Seagate is advertising servo architecture roles including AI/ML-in-control. Servo demand should stay solid, though not grow fast, because only three companies employ it.

5. Opportunities and risks, 2026–2031

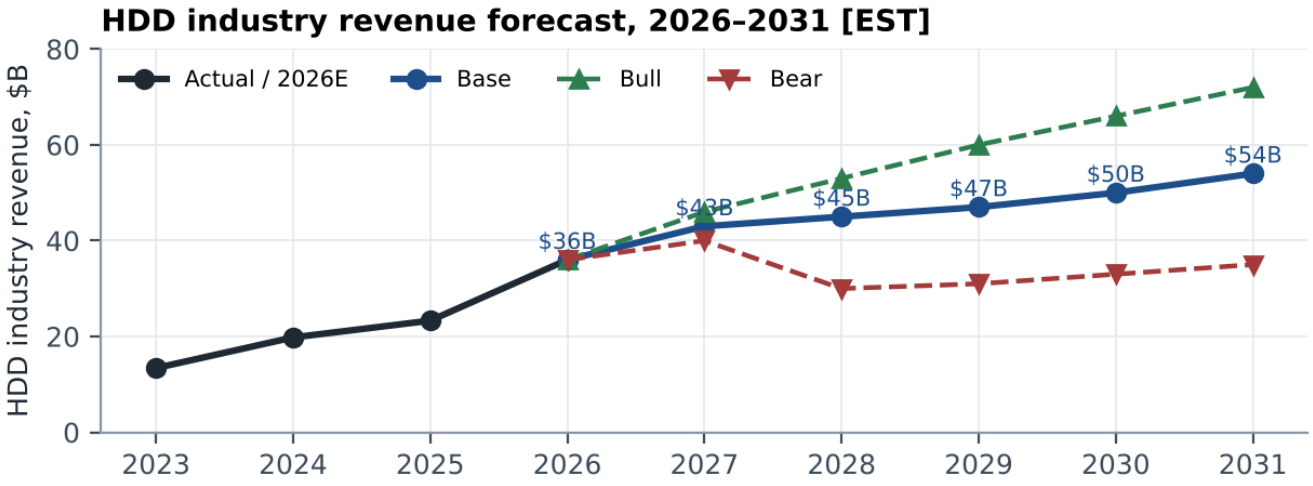
Opportunity	Why it matters	Likelihood
AI data retention	Inference/agentic AI creates persistent data; vendors target >25%/yr exabyte growth	High
Pricing power	3 suppliers, no greenfield capacity, LTAs with value-based pricing	High to 2027; medium after
Areal-density step-up (HAMR)	~25% exabyte growth with flat units and low capex → high incremental margins	High
Broader customer base	Neoclouds, AI labs, sovereign clouds, enterprise	Medium-high
NAND shortage	Keeps SSD 13–22x more expensive per TB in 2026–27	High to 2027, fades 2028+

Risk	Mechanism	Impact	Probability (5 yr)
AI capex digestion / bubble	Capex outruns cash flow (Meta FCF $8.6B → $0.8B/qtr); 2–4 qtr lag to HDD, cushioned by LTAs	High	Medium (40–50%)
Price normalization	Toshiba doubling capacity; HAMR adds ~40% TB/drive; $/TB returns to −5–10%/yr after ~2028	High on margins	High
NAND glut → QLC substitution	New fabs 2028–29; gap back to 3–5x	Medium	Medium
Customer concentration	Cloud ~88% of WD revenue; top-10 = 73%	Medium	Ongoing
HAMR execution	WD/Toshiba slips change share, not total demand	Company-level	Medium
Power constraints	SSD watts/TB argument gains weight in power-limited DCs	Low-medium	Medium
Tariffs / China	Assembly in Thailand, Malaysia, Philippines, China	Low-medium	Medium
Tape / new media	Cold-archive tier only	Low	Low

History is the best guide to downside: the 2023 cloud inventory correction cut industry revenue ~40% (to ~$13.4B) and pushed gross margins below 25%. The industry is better protected now (contracts, lean opex, net cash), but a similar-shaped correction in 2027–29 is the main scenario to plan around.

6. Outlook: does HDD still have growth? How much?

Yes, in capacity; probably yes in value, but more slowly and with a dip. The forecast below is manufacturer revenue (comparable to Coughlin Associates' definition). Base case assumes exabytes grow ~20%/yr from 2.05 ZB to ~5 ZB (between Furthur's 4.7 ZB in 2031 and Coughlin's 6.5 ZB in 2030), price per TB is flat through 2027 then falls ~10%/yr, and one digestion year occurs in 2028.



Scenario	2027	2028	2029	2030	2031	CAGR 2026–31	Assumptions
Bull	$46B	$53B	$60B	$66B	$72B	~15%	Pricing discipline holds; EB +25%/yr; QLC stays expensive
Base	**$43B**	**$45B**	**$47B**	**$50B**	**$54B**	**~8%**	EB ~20%/yr to ~5 ZB; $/TB −10%/yr from 2028; one soft year
Bear	$40B	$30B	$31B	$33B	$35B	~0%	AI capex correction 2027–28; $/TB −15%/yr; QLC takes 10–15% of nearline EB

Other published forecasts	Figure	Comment
Coughlin Associates (Dec 2025)	$25–35B revenue by 2030; 6.5 ZB	Revenue already exceeded in 2026; needs upward revision
Furthur Market Research (Mar 2026)	HDD peaks ~4.7 ZB/yr in 2031; end-of-life ~2043	Long tail: HDD still significant into late 2030s
Micron (FMS 2025)	HDD storage bits +26% CAGR 2024–2030	Even a flash maker expects HDD growth
Mordor Intelligence	$51.8B (2026) → $75.6B (2032), 6.6% CAGR	Broader end-market definition (retail value)

Assessment. HDD has a durable 5-year growth runway in exabytes (~20%/yr) underpinned by AI data retention, a $/TB advantage that no competitor closes before 2031, and contracts into 2028–31. Value growth is likely ~8%/yr on average but uneven: 2026–27 is a peak-pricing period, and a 2028-ish price/margin reset is more likely than not. The industry's structural end-point (Furthur: ~2040s) is beyond a 10-year horizon, which is relevant for career planning: **HDD remains a viable industry for at least the next 10–15 years, but plan for one downturn with layoffs in the next 5.**

Sources

- Seagate FQ1 FY2027 press release (SEC 8-K, 27 Oct 2026): https://www.sec.gov/Archives/edgar/data/0001137789/000113778926000300/stxq12027pressreleasefinan.htm
- Western Digital FQ1 FY2027 press release (SEC 8-K, 22 Oct 2026): https://www.sec.gov/Archives/edgar/data/0000106040/000162828026066882/a4ex991-pressreleaseq127.htm
- Western Digital FQ4 FY2026 press release (SEC 8-K, 5 Aug 2026): https://www.sec.gov/Archives/edgar/data/0000106040/000162828026053305/a4ex991-pressreleaseq426.htm
- Tom Coughlin, C4Q 2025 and 2025 HDD industry update, Forbes (2 Feb 2026): https://www.forbes.com/sites/tomcoughlin/2026/02/02/c4q-2025-and-2025-hard-disk-drive-industry-update/
- Tom Coughlin, Digital storage projections for 2026, Forbes (19 Dec 2025): https://www.forbes.com/sites/tomcoughlin/2025/12/19/digital-storage-and-memory-projections-for-2026-part-1/
- Gartner, IT spending forecast (27 Jul 2026): https://www.gartner.com/en/newsroom/press-releases/2026-07-27-gartner-forecasts-worldwide-it-spending-to-grow-14-point-2-percent-in-2026-totaling-6-point-37-trillion
- Platformonomics, Follow the CAPEX Q2 2026: https://platformonomics.com/2026/07/follow-the-capex-q2-2026-scoreboard/
- DatacenterDynamics, Moody's hyperscaler capex near $1T by 2027: https://datacenterdynamics.com/en/news/moodys-hyperscaler-capex-forecasts-marked-up-by-85bn-to-close-in-on-1trn-by-2027/
- 247wallst, Citi $5.35T hyperscaler capex 2027–2030 (21 Oct 2026): https://247wallst.com/investing/2026/10/21/citi-sees-amazon-microsoft-google-and-meta-spending-more-than-apple-is-worth-on-ai-by-2030/
- TrendForce, 2Q26 eSSD ranking: https://www.trendforce.com/presscenter/news/20260901-13210.html ; 2Q26 NAND ranking: https://www.trendforce.com/presscenter/news/20260818-13186.html
- TrendForce, 2026 memory capex: https://www.trendforce.com/presscenter/news/20251113-12780.html
- TrendForce, Toshiba to double AI DC HDD capacity by FY2027: https://trendforce.com/news/2026/10/02/news-toshiba-to-double-ai-data-center-hdd-capacity-by-fy2027-with-380m-expansion-as-ai-storage-demand-surges/
- Micron FQ4 FY2026 press release: https://www.sec.gov/Archives/edgar/data/0000723125/000072312526000018/a2026q4ex991-pressrelease.htm
- Sandisk FQ4 FY2026 results: https://www.sandisk.com/company/newsroom/press-releases/2026/2026-08-05-sandisk-reports-fiscal-fourth-quarter-2026-financial-results
- Blocks & Files / VDURA, QLC 22.6x HDD: https://www.blocksandfiles.com/flash/2026/04/08/vdura-says-30-tb-qlc-ssd-capacity-now-costs-226x-more-than-hdd/5214761
- Notebookcheck, enterprise SSDs 18.6x HDD (Aug 2026): https://www.notebookcheck.net/Enterprise-SSDs-now-cost-18-6x-more-than-HDDs-prices-soar-6-5x-in-a-year.1374822.0.html
- John Monroe (Furthur), Library of Congress DSA (Mar 2026): https://www.digitalpreservation.gov/meetings/DSA2026/0107_monroe_Library%20of%20Congress_DSA_09%20March%202026%20V11.83-Final.pdf
- Micron, T. Grossi, FMS 2025: https://files.futurememorystorage.com/proceedings/2025/20250807_AIML-302-1_T.Grossi.pdf
- Blocks & Files, Everpure Q2 FY27: https://www.blocksandfiles.com/flash/2026/08/27/everpure-the-artist-formerly-known-as-pure-storage-ramps-up-growth-rate/5292871
- Blocks & Files, Seagate HAMR 40% of nearline EB (30 Jul 2026): https://www.blocksandfiles.com/disk/2026/07/30/seagate-hamring-customers-generating-more-and-more-/tb/5281058
- Tom's Hardware, Seagate 44TB Mozaic 4+ shipping: https://tomshardware.com/pc-components/hdds/seagate-begins-shipping-44tb-hard-drives-with-hamr-tech-to-data-centers-mozaic-4-platform-expands-to-10-platters
- StorageReview, WD 40TB UltraSMR, HBD, Dual Pivot, 100TB roadmap: https://www.storagereview.com/news/wd-bets-8x-bandwidth-beats-more-terabytes-as-40tb-ultrasmr-starts-shipping
- NAND Research, WD Innovation Day note (Feb 2026): https://nand-research.com/research-note-western-digital-innovation-day/
- TechRadar, Toshiba roadmap to 55TB: https://www.techradar.com/pro/toshiba-wants-to-launch-a-55tb-hard-drive-by-2030-40tb-model-set-to-appear-in-2026-new-slides-show
- TrendForce, race to 400-layer NAND: https://www.trendforce.com/news/2026/06/12/news-the-race-to-400-layer-nand-roadmaps-and-key-technologies-driving-samsung-sk-hynix-and-kioxia/
- Blocks & Files, LTO roadmap cut: https://www.blocksandfiles.com/tape/2025/11/13/lto-10-bumped-to-40-tb-as-future-tape-capacities-get-cut/1711787 ; tape 2025/2026: https://www.blocksandfiles.com/tape/2026/09/04/tape-capacity-shipments-fell-in-2025-but-2026-could-be-a-boom-year-for-tape/5294464
- Blocks & Files, Project Silica: https://blocksandfiles.com/2025/12/23/project-silicas-glass-storage-archive-tech-progress/ ; Cerabyte roadmap: https://www.blocksandfiles.com/data-management/2025/07/02/cerabyte-promises-fast-dense-low-cost-tape-alternative-by-2030/1595617
- Mordor Intelligence, HDD market: https://mordorintelligence.com/industry-reports/hard-disk-drive-market
- Glenn K. Lockwood, 2025–26 storage shortage notes: https://glennklockwood.com/garden/shortage